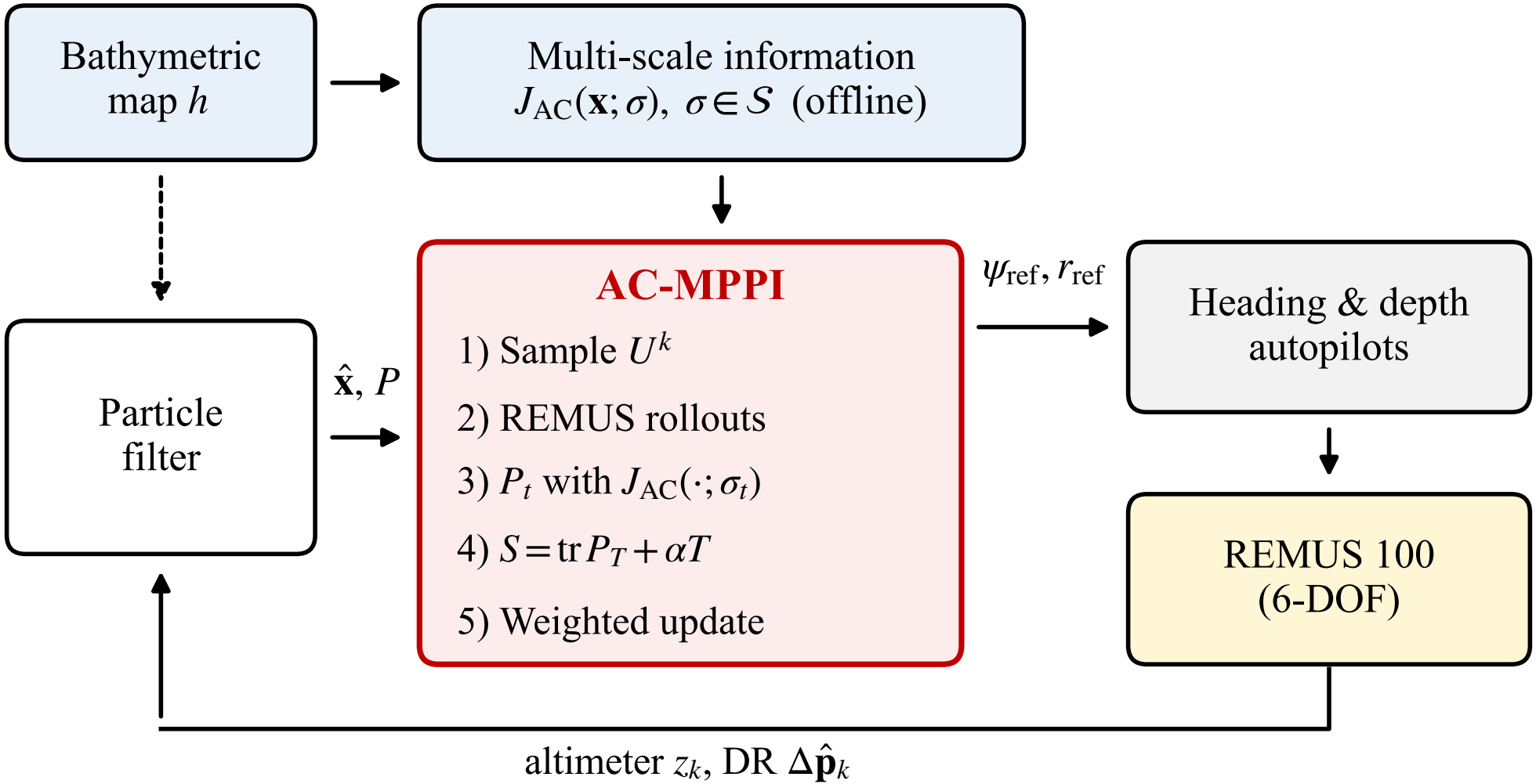

Bathymetric map h
Multi-scale information $J_{\mathrm{AC}}(\mathbf{x};\sigma),\ \sigma \in S$ (offline)
Particle filter
$\hat{\mathbf{x}}$, P
AC-MPPI
1) Sample U^k
2) REMUS rollouts
3) P_t with $J_{\mathrm{AC}}(\cdot;\sigma_t)$
4) $S = \mathrm{tr}\, P_T + \alpha T$
5) Weighted update
ψ_{ref}, r_{ref}
Heading & depth autopilots
REMUS 100 (6-DOF)
altimeter z_k, DR $\Delta\hat{\mathbf{p}}_k$